

ABSTRACT ALGEBRA

HOMEWORK 3

Please hand in your solutions to the following problems by 2025-08-01. The homework has to be submitted via the course website as a single pdf file. It has to be typeset; handwritten homework will not be graded as the platform won't accept it. TeX is strongly preferred, but we will accept homework typeset with MS Word or similar, so long as the text is easy to read and meets reasonable standards of mathematical writing. You are encouraged to discuss the problems with your colleagues, but must hand in your own solutions. You are welcome to ask us questions, especially in connection to the discussion session.

Problem 1. (5 points) Let

$$\mathbb{H} = \{a \cdot 1 + b \cdot i + c \cdot j + d \cdot k \mid a, b, c, d \in \mathbb{R}, i^2 = j^2 = k^2 = -1, ij = -ji = k\}$$

endowed with the following laws:

- for any $a, a', b, b', c, c', d, d' \in \mathbb{R}$ one has $(a \cdot 1 + b \cdot i + c \cdot j + d \cdot k) + (a' \cdot 1 + b' \cdot i + c' \cdot j + d' \cdot k) = (a + a') \cdot 1 + (b + b') \cdot i + (c + c') \cdot j + (d + d') \cdot k$;
- a multiplication defined by the relation for i, j, k (in the definition of $\mathbb{H}$), such that 1 is the neutral element for the multiplication and it is associative and bilinear.

An element of $\mathbb{H}$ is called a quaternion.

- (1) Show that $\mathbb{H}$ is a non-commutative ring. **(1 point)**
- (2) For any $x = a \cdot 1 + b \cdot i + c \cdot j + d \cdot k \in \mathbb{H}$ set $\bar{x} := a \cdot 1 - b \cdot i - c \cdot j - d \cdot k$ and define the norm of x as $N(x) = x\bar{x}$. Show that:
 - (a) $N(xy) = N(x)N(y)$ for any $x, y \in \mathbb{H}$,
 - (b) $N(x) = 0$ if and only if $x = 0$.**(1 point)**
- (3) Show that $\mathbb{H}$ has no zero-divisors and that any non-zero element of $\mathbb{H}$ is invertible. One says that $\mathbb{H}$ is a non-commutative field. **(1 point)**
- (4) Let now define the Hamilton quaternions as the set $H = \mathbb{Z} \cdot i + \mathbb{Z} \cdot j + \mathbb{Z} \cdot k + \mathbb{Z} \cdot 1$ with $w = \frac{1}{2}(1 + i + j + k)$. Show that H is a subring of $\mathbb{H}$. Does it have zero divisors? **(1 point)**
- (5) Let A be a ring with no zero-divisor and $\phi : A \rightarrow \mathbb{N}$ a map. The ring A is said to be right euclidean (respectively left euclidean) with respect to ϕ if for any $a, b \in A$ with $b \neq 0$ there exist $q, r \in A$ such that $a = qb + r$ (respectively $a = bq + r$) and either $r = 0$ or $\phi(r) < \phi(b)$.
Show that H is a left and right euclidean ring with respect to the norm N restricted to H . **(1 point)**

Problem 2. (2 points) Let K be a field, a splitting field of a polynomial $f(X)$ over K is a field extension $L = K[a_1, \dots, a_n]$ of K where:

- the a_i 's are roots of $f(X)$
- and $f(X) = c \prod_{i=1}^{\deg(f)} (X - a_i)$ where $c \in K$, is the leading coefficient of $f(X)$.

Let $f(x) = x^4 - 5x^2 + 6 \in \mathbb{Q}[x]$ and E be a splitting field of the polynomial f .

- (1) Show that $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and determine $[E : \mathbb{Q}]$ the degree of the field extension. **(1 point)**.
- (2) Show that $E = \mathbb{Q}(\sqrt{2} + \sqrt{3})$. **(1 point)**.

Problem 3. (1 point)

Let $K = \mathbb{F}_3[X]/(X^2 + 1)$. Determine a polynomial $g(x) \in \mathbb{F}_3[X]$ such that $g(X) + (X^2 + 1)$ is a generator of K^* , the group of invertible elements of K .

Problem 4. (4 points)

A ring A is Boolean if $x^2 = x$ for all $x \in A$. In a Boolean ring A show that:

- (1) For all $x \in A$ one has $2x = 0$, **(1 point)**
- (2) Every prime ideal $\mathfrak{p}$ is maximal and $A/\mathfrak{p}$ is a field with two elements, **(1 point)**
- (3) Every finitely generated ideal in A is principal, **(1 point)**
- (4) Is there a Boolean ring which is not a principal ideal domain? **(1 point)**

Problem 5. (3 points)

Let A be a unique factorization domain, and let $F \in A[X]$. The polynomial F is primitive if the only elements $a \in A$ that divide all coefficient of f are units.

- (1) Show that if $S, T \in A[X]$ are primitive so is their product $ST \in A[X]$. **(1 point)**
- (2) Deduce that if A is a unique factorization domain, so is $A[X]$. **(2 points)**

Problem 6. (5 points) Let I be an ideal of $\mathbb{Z}[X]$, the polynomial ring with integer coefficients, such that:

- (1) elements of I do not have a common divisor of degree greater than 0,
- (2) I contains a constant polynomial with constant term 1.

Prove that I contains the polynomial $1 + x + x^2 + \cdots + x^r$ for some natural number r .

Hint:

- Show that there exists a polynomial $f \in \mathbb{Z}[X]$ such that $(1 + xf) \in I$.
- Show that $x^r \in \langle 1 + xf, x^{r+1} \rangle$, the ideal generated by $1 + xf$ and x^{r+1} .
- Decompose $1 + x + x^2 + \cdots + x^r$ as $(1 + xf)a_r + x^{r+1}b_r$ with $a_r, b_r \in \mathbb{Z}[X]$ the idea is to show that there exists an $r \in \mathbb{N}$ such that $b_r \in I$. To do so, show that we can always choose b_r in such a way that $\deg(b_r) \leq \deg(f)$. Then show that I contains a non zero constant polynomial δ . Show that there exists $s \in \mathbb{N}$, $s > r$ such that $b_s - b_r \in \langle \delta \rangle$.